No multiplicative persistence 12 below 10^{40000}: a sieve over chain numbers

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Abstract

Let $p(n)$ be the product of the decimal digits of n. The multiplicative persistence of n is the number of applications of p needed to reach a single digit. The record value 11 is attained by 277777788888899, and it is conjectured that no number has persistence 12. The best published lower bound for the smallest number of persistence 12, if it exists, was $2.67 \cdot 10^{30000}$ (T. Peters, 2023). We show that no number below 10^{40000} has persistence 12 or more, so the smallest one, if it exists, exceeds $2.67 \cdot 10^{40000}$. The proof reduces the question to *chain numbers*: 7-smooth numbers with no digit 0 that do not contain both a 5 and an even digit. Measured by a "digit length" adapted to the digit product, every chain number up to length 40,000 is one of 1,117 known numbers, all below 10^{49}. The search covers $6.4067 \cdot 10^{13}$ candidates. It rests on four elementary lemmas: a suffix collapse that turns the last 13 digits of every candidate into one bit of a precomputed bitmap, an exact 4^k count of admissible endings, an affine formula for the next 13 digits, and the uniqueness of an odd-digit 2-adic "thread" for multiples of high powers of 5. We describe the implementation and its validation: exact-arithmetic comparisons, an independent reference implementation, rediscovery of known zero-free numbers, and two complete re-executions on different hardware with an independent implementation of the core stages. A random-digit heuristic, calibrated against the complete data, is reported separately; it is not part of the proof.

1 Introduction

For a positive integer n let $p(n)$ be the product of its decimal digits and let $\mathrm{per}(n)$ be the least $k \geq 0$ with $p^k(n) < 10$. The smallest numbers of persistence $0, 1, \ldots, 11$ form OEIS A003001 [1], ending with

$$277777788888899 \to 4996238671872 \to 438939648 \to 4478976 \to 338688 \to 27648$$
$$\to 2688 \to 768 \to 336 \to 54 \to 20 \to 0.$$

Sloane [2] asked whether persistence is bounded; the standing conjecture is that 11 is the maximum in base 10. Orbits that end in an odd digit are settled by Brier, Clavier, Gutsche and Naccache [3]; a 2-adic bound of Fonga [4] makes the nonzero even final digits a finite (if large) computation. The hard case is the final digit 0, which is where the record orbit ends: it amounts to understanding the 7-smooth numbers without a digit 0, a problem of the same family as Erdős's question whether 2^{86} is the last power of 2 without a digit 0.

*Computations and drafting were done with the assistance of an AI system (Claude, Anthropic). Code, logs and validation records: `https://github.com/CorvusCorpus/persistence-research`.

Lower bounds for the smallest number $a(12)$ of persistence 12 come from computation. Chaffin [5] found that every 7-smooth number in $(10^{140}, 10^{20000})$ contains a digit 0, which gives $a(12) > 10^{20585}$ (Radcliffe [6]). Peters [7] enumerated the smallest representative of every digit-product class up to 30,000 digits (about $3.6 \cdot 10^{13}$ candidates) and checked that each digit product has a 0 among its last 306 digits, giving $a(12) > 2.67 \cdot 10^{30000}$. We prove:

Theorem 1. *No positive integer below 10^{40000} has multiplicative persistence 12 or more. Consequently, if $a(12)$ exists, then $a(12) > 2.67 \cdot 10^{40000}$.*

The proof is computer-assisted (Section 6). Its new ingredient is a sieve over *chain numbers* (Section 2) in which most candidates are rejected by a single bit lookup that tests their last 13 digits (Section 3). Section 5 describes how the computation was checked, Section 7 what a counterexample above the bound would have to look like, and Section 8 a heuristic that is kept apart from the proof.

2 Chain numbers

Definition 2. A number is 7-*smooth* if all its prime factors are at most 7. For $x = 2^a 3^b 5^d 7^c$ put

$$m(x) = \tfrac{a}{3} + \tfrac{b}{2} + c + d,$$

the *digit length* of x. A *chain number* is a 7-smooth $x \geq 10$ with no digit 0 that does not contain both a digit 5 and an even digit. We write G for the set of chain numbers.

A chain number is never divisible by 10, so it has one of the forms $2^a 3^b 7^c$ (form (i)) or $3^a 5^b 7^c$ with $b \geq 1$ (form (iii)); in form (iii) the last digit is 5, so every digit is odd. (Splitting the search for zero-free digit products into these two families is standard; see [8].)

Lemma 3 (chain lemma)**.** *Let n have persistence $k \geq 12$ and put $c_i = p^{i+1}(n)$. Then $c_0, \ldots, c_8$ are chain numbers, and $m(c_0) \leq \mathrm{len}(n)$, the number of digits of n.*

Proof. By definition $c_i \geq 10$ for $i \leq k-2$, and c_{k-1} is a single digit. Every c_i is a digit product, hence 7-smooth, and $c_i \geq 10$ is nonzero. For $i \leq k-3$, $c_{i+1} = p(c_i) \neq 0$, so c_i has no digit 0. If c_i contained a 5 and an even digit, then $10 \mid p(c_i) = c_{i+1}$, so c_{i+1} would end in 0 and $c_{i+2} = 0$, which is impossible for $i+2 \leq k-2$. Hence $c_i \in G$ for $i \leq k-4$, in particular for $i \leq 8$. For the second claim, a digit d of n multiplies $p(n)$ by d, which raises m by $0, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, 1, \frac{5}{6}, 1, 1, 1$ for $d = 1, \ldots, 9$; each contribution is at most 1. □

Corollary 4. *Suppose every chain number x with $m(x) \leq N$ is one of a known finite list, all of whose members have persistence at most 10. Then every n with at most N digits has persistence at most 11.*

Proof. If $\mathrm{per}(n) \geq 12$ and $\mathrm{len}(n) \leq N$, then $c_0 = p(n)$ is a chain number with $m(c_0) \leq N$ (Lemma 3), hence on the list, and $\mathrm{per}(n) = 1 + \mathrm{per}(c_0) \leq 11$. □

The bound $a(12) > 2.67 \cdot 10^N$ follows from Corollary 4 by the candidate argument of Peters [7]. Suppose $a = a(12)$ exists. Permuting digits, deleting a digit 1, or replacing some digits by fewer digits with the same product ($22 \to 4$, $23 \to 6$, $24 \to 8$, $33 \to 9$, $266 \to 89$) gives a number with the same digit product, hence the same persistence, and a smaller one unless nothing changes; and a number containing a 5 and an even digit has persistence at most 2. So the digits of a are nondecreasing and contain none of these configurations. By Corollary 4, a has at least $N+1$ digits. If its first digit is 2, its second is at least 6, and after $2, 6$ every further digit is at least 7. Hence $a \geq 2\,6\,7 \cdots 7$ (with $N+1$ digits) $> 2.67 \cdot 10^N$.

Remark 5. A chain number with a 5 has only odd digits. Per digit, a random number satisfies the chain condition with probability about 0.8, against 0.9 for being zero-free; this, and the weight m, which charges each prime by the shortest digit string that produces it, are what make chain numbers much rarer than zero-free 7-smooth numbers: 1,117 chain numbers are known, the largest with 49 digits, against 12,606 zero-free ones up to 140 digits.

3 Suffix structure

Throughout, $K \geq 1$, $P_K = 4 \cdot 5^{K-1}$ is the order of 2 modulo 5^K, and μ_3, μ_7 are the discrete logarithms $2^{\mu_3} \equiv 3$, $2^{\mu_7} \equiv 7 \pmod{5^K}$ (they exist because 2 is a primitive root modulo every power of 5).

Lemma 6 (suffix collapse)**.** *If $a \geq K$ then $2^a 3^b 7^c \equiv 2^e \pmod{10^K}$ for every $e \geq K$ with $e \equiv a + b\mu_3 + c\mu_7 \pmod{P_K}$.*

Proof. Both sides are $\equiv 0 \pmod{2^K}$, and both are $\equiv 2^{a+b\mu_3+c\mu_7} \pmod{5^K}$; apply the Chinese remainder theorem. □

So the last K digits of every form-(i) number with $a \geq K$ are those of a power of two, determined by one residue modulo P_K. For $K = 13$, a bitmap of $P_{13} = 976{,}562{,}500$ bits records which residues have last 13 digits free of 0 and 5. Along a *row* (b, c fixed, a increasing) the residue increases by one at each step, so a row is a contiguous range of bits.

Lemma 7 (exact branching)**.** *For $k \geq 1$, among the P_k residues $e \bmod P_k$ ($e \geq k$), exactly 4^k give a power 2^e whose last k digits avoid 0 and 5.*

Proof. For $k = 1$ the last digit of 2^e runs through $2, 4, 8, 6$. Fix $e \geq k + 1$ and consider the five lifts $e_t = e + tP_k$, $t = 0, \ldots, 4$, of $e \bmod P_k$ to residues modulo P_{k+1}. Write $2^{e_t} \equiv x_0 + 10^k d_t \pmod{10^{k+1}}$ with digits d_t; the last k digits do not depend on t. Since $2^{P_k} = 1 + 5^k u$ with $5 \nmid u$, we have $2^{e_t} \equiv 2^e(1 + t\,5^k u) \pmod{5^{k+1}}$, so $2^k(d_t - d_0) \equiv 2^e t u \pmod 5$ and the d_t are distinct modulo 5. Modulo 2^{k+1} both 2^{e_t} and 2^{e_0} vanish, so $2^k 5^k (d_t - d_0) \equiv 0$ and the d_t have equal parity. Hence the d_t are the five digits of one parity, and exactly one of them lies in $\{0, 5\}$. □

The bitmap therefore has exactly $4^{13} = 67{,}108{,}864$ set bits (a density of $0.8^{12} \approx 0.0687$), and stage 1 of the sieve passes exactly that fraction of a complete period. Lifting arguments of this kind are standard for digit problems on powers of two (see, e.g., [9]); the exact count for the excluded set $\{0, 5\}$ is what makes the chain condition well suited to a bitmap.

Lemma 8 (affine lift)**.** *Let $Q = 5^K$, $P = P_K$, $0 \leq e < P$, and let $y = 2^{e-2K} \bmod 5^{2K} = A_{\mathrm{lo}} + Q A_{\mathrm{hi}}$ with $0 \leq A_{\mathrm{lo}}, A_{\mathrm{hi}} < Q$. Let $w = (2^P \bmod 5^{2K} - 1)/Q$. Then for every $j \geq 0$ with $e + jP \geq 2K$, the digits $K + 1, \ldots, 2K$ of 2^{e+jP} form the integer*

$$\left\lfloor \frac{2^K A_{\mathrm{lo}}}{Q} \right\rfloor + 2^K \big((A_{\mathrm{hi}} + A_{\mathrm{lo}}\, w\, j) \bmod Q \big).$$

Proof. $2^{e+jP} \bmod 10^{2K} = 2^{2K}\big(2^{e+jP-2K} \bmod 5^{2K}\big)$, and $2^{e+jP-2K} \equiv y(1 + Qw)^j \equiv y(1 + Qwj) \equiv A_{\mathrm{lo}} + Q(A_{\mathrm{hi}} + A_{\mathrm{lo}} w j) \pmod{Q^2}$. Dividing $2^{2K}(A_{\mathrm{lo}} + QY)$, $Y < Q$, by $10^K = 2^K Q$ gives the stated quotient. □

By Lemma 6 with K and $2K$, the full exponent residue of a form-(i) number modulo P_{2K} is $e + jP_K$ with j constant along a row. So after the bitmap, the next K digits of each surviving point cost one modular multiplication by the row constant j.

Lemma 9 (odd thread)**.** *There is exactly one residue r_{40} modulo 10^{40} with $5^{40} \mid r_{40}$ and all 40 digits odd:*

$$r_{40} = \cdots 9313975971735999337397515773773193359375.$$

Write $r_{40} \equiv -5^t \pmod{2^{40}}$ and $3 \equiv -5^{f_3}$, $7 \equiv -5^{f_7} \pmod{2^{40}}$. If $b \geq 40$, the last 40 digits of $x = 3^a 5^b 7^c$ are all odd if and only if $a + c$ is odd and $b \equiv t - af_3 - cf_7 \pmod{2^{38}}$.

Proof. If r has k odd digits and $5^k \mid r$, the lifts $r + j10^k$ divisible by 5^{k+1} are $j \equiv j_0 \pmod 5$, i.e. j_0 and $j_0 + 5$, one odd and one even; so exactly one lift survives, starting from $r = 5$. For $b \geq 40$, $x \equiv 0 \pmod{5^{40}}$ and $x \equiv (-1)^{a+c} 5^{\,b+af_3+cf_7} \pmod{2^{40}}$; since 5 has order 2^{38} modulo 2^{40} and -1 is not a power of 5 there, $x \equiv r_{40} \pmod{10^{40}}$ exactly under the stated conditions. □

So for each pair (a, c) at most one residue class of b modulo 2^{38} survives the last 40 digits: the odd form costs $O(1)$ per pair (a, c) for $b \geq 40$.

4 The sieve

Regions. By Corollary 4 it suffices to examine every $x = 2^a 3^b 7^c$ with $2a + 3b + 6c \leq 6N$ and every $x = 3^a 5^b 7^c$, $b \geq 1$, with $a + 2b + 2c \leq 2N$. All such $x < 10^{131}$ (there are 4,483,467) were enumerated directly; among them are exactly the 1,117 chain numbers of the known list [12]. The sieve handles $x \geq 10^{130}$, so the two ranges overlap. Shells $N_0 < m \leq N$ are run separately with an extra lower bound.

Main path (form (i), $a \geq 26$, $x \geq 10^{130}$). Each candidate passes through

1. the bitmap of Lemma 6, $K = 13$ (last 13 digits): survival 0.8^{12};
2. the affine lift of Lemma 8 (digits 14–26): survival 0.8^{13} of the rest;
3. a leading-digit test: $t = aL_2 + bL_3 + cL_7 \bmod 2^{64}$ with $L_p = \mathrm{round}(2^{64} \log_{10} p)$ approximates $2^{64}\{\log_{10} x\}$ within $(a + b + c)/2$ units, and a 2^{18}-bin table rejects only bins in which every point (with a margin of 2^{-40}) has a 0 or 5 among its first 5 significant digits;
4. exact digits 27–126 by a base-10^9 limb product;
5. GMP: the last 600 digits, then the whole number.

Only stage 3 uses approximations, and it rejects a candidate only when the rejection is certain under a proved error bound; everything else is exact integer arithmetic.

Direct and odd paths. Form (i) with $a < 26$ is stepped along b with three 18-digit limbs. Form (iii) with $b \leq 39$ is stepped along a with an 18-digit filter, and $b \geq 40$ uses Lemma 9. Survivors are decided with GMP.

Cost. The region has about N^3 points, so the cost grows like N^3. Stages 1–2 cost about 0.17 ns per candidate on one modern CPU core, or about 5 ps on an RTX 3060 Ti.

5 Implementation and validation

The program (`gsieve.c`, C with GMP, about 1,000 lines) has several interchangeable implementations of stages 1–2: AVX-512 with IFMA (the original production build), AVX-512, AVX2, a scalar reference, and CUDA.

- *Arithmetic.* Every vector implementation divides by 10^4 as $\lfloor vM/2^{45} \rfloor$ with $M = \lceil 2^{45}/10^4 \rceil$. This is exact for $0 \le v < 2^{45}/1168$ (Granlund–Montgomery [10]), and the operands are split so that every divided value is below 2^{31}; the identity was checked for all $v < 2^{31}$. The lift uses Montgomery multiplication [11] modulo 5^{13}.
- *Mutual agreement.* All implementations make identical decisions. Dumps of every stage-2 survivor and stage-3/4 decision agree across implementations and thread counts, and the CUDA dumps are byte-identical to the CPU dumps.

The computation was checked as follows.

1. *Exact arithmetic on full regions.* Every stage decision on full regions up to $N = 150$ (both $K = 12$ and $K = 13$ builds) and on complete production slices (single values of c at $N = 36{,}000$ and in the shell, up to $3.6 \cdot 10^8$ candidates each) was compared with big-integer arithmetic.
2. *Positive control.* In a zero-free mode the sieve must rediscover known numbers. End to end at $N = 160$ it reproduces exactly the 314 zero-free 7-smooth numbers of that size from an independent list complete below 10^{5000}.
3. *Brute force.* The direct and odd paths agree with brute force on full small regions. The odd thread was also checked digit by digit on production regions that contain real thread candidates.
4. *Independent reference.* A separate plain-Python implementation shares no code with the program and decides everything from the digits of x; it re-derives the constants, Lemma 7 for $k \le 9$ and the odd thread. It agrees with the program on full regions and on production slices, including all odd-thread candidates found there.
5. *Kernel tests (GPU).* On the whole table, $6.9 \cdot 10^{10}$ stage-2 decisions of the GPU agree with an independent host formula. On 2^{20} random points, the decisions agree with $2^E \bmod 10^{26}$ computed by GMP.
6. *Complete re-executions.* The logged production run ($6.4067 \cdot 10^{13}$ candidates, computed on a cloud machine with ECC memory) was re-executed twice on a desktop PC with the GPU implementation of stages 1–2. Every counter of every invocation and every per-c count agreed with the original logs.
7. *Statistics.* Over the full runs the survival rates of stages 1 and 2 are 0.06872 and 0.05497, against $0.8^{12} = 0.068719$ and $0.8^{13} = 0.054976$.
8. *Fault tolerance.* On machines without ECC memory, the GPU tables are re-verified every minute against host copies, and output is written only after a passed check. New shells also get an independent second pass on the CPU, compared counter by counter.

run	main path	direct	odd	total
$m \le 36{,}000$	46,626,195,781,376	33,689,253,108	50,490,005,607	46,710,375,040,091
$36{,}000 < m \le 40{,}000$	17,337,008,704,000	7,903,308,000	11,850,072,000	17,356,762,084,000
$m \le 40{,}000$				64,067,137,124,091

Table 1: Candidates examined (plus 4,483,467 numbers below 10^{131} enumerated directly). No run produced a chain number.

6 Results

Theorem 10 (computational)**.** *Every chain number x with $m(x) \le 40{,}000$ is one of the* 1,117 *chain numbers below* 10^{49}*; the largest is* $461382462366246987746371894149916684998327612211 2 = 2^{132} \cdot 3 \cdot 7^{10}$*. All of them have persistence at most* 10*, and only* $4996238671872 = 2^{19}3^47^6$ *and* $937638166841712 = 2^43^{20}7^5$ *have persistence exactly* 10*.*

Theorem 1 follows from Theorem 10 and Corollary 4.

7 Above the bound: bridges

Lemma 11. *Assume Theorem 10 for $N = 40{,}000$, and let n have persistence at least* 12*, with c_i as in Lemma 3. Then either $c_0, \dots, c_8$ all have $m > 40{,}000$, or some c_t ($t \le 7$) is a* bridge*: a chain number with more than* 27,958 *digits whose digit product is one of the* 1,117 *known chain numbers. A bridge has fewer than* 162 *digits different from* 1*, so more than* 99.4% *of its digits are* 1*s.*

Proof. By Theorem 10, $m(c_0) > 40{,}000$; let t be the largest index ≤ 8 with $m(c_t) > 40{,}000$. If $t < 8$ then $c_{t+1} = p(c_t)$ is a chain number with $m \le 40{,}000$, hence known and below 2^{162}, and each digit of c_t other than 1 contributes a factor at least 2. Finally $m(x) \le \log_{10} x / \log_{10} 5 < 1.431 \log_{10} x$ for 7-smooth x, so $m(c_t) > 40{,}000$ forces more than 27,958 digits. □

A bridge is a near-repunit: writing it as $(10^L - 1)/9 + \sum_i (d_i - 1)10^{a_i}$ with at most 161 digits $d_i \ne 1$ gives S-unit equations with up to 161 terms, far beyond the reach of the linear-forms methods of [3, 4].

8 A heuristic (not part of the proof)

Model the digits of a 7-smooth number with L digits as independent and uniform. Against the complete list below 10^{5000} the model predicts 5,622 zero-free numbers with 20–39 digits (actual 5,581), 2,059 with 40–59 digits (actual 2,111), and 7.3 chain numbers with 40–59 digits (actual 7). A number of persistence 12 exists if and only if some 7-smooth number has digit product 4996238671872 or 937638166841712 (unless an unknown persistence-10 digit product exists beyond 10^{20000}). Under the model, the expected number of such 7-smooth numbers is about 0.019, concentrated at 18–22 digits, a range long excluded. The expected number of chain numbers beyond 25,000 digits is below 10^{-2400}. Larger computational bounds therefore certify an exponentially thin tail; they are not evidence in either direction about the conjecture.

9 Reproducibility

The repository [12] contains the source, the production logs (∼13 MB, plus compressed archives for later shells), SHA-256 manifests, and all validation logs with the exact commands, build flags, binary hashes and commits. The production of the original run took about 5.5 core-hours on a 2.1 GHz AVX-512 machine; the complete re-execution takes about 8 minutes on an RTX 3060 Ti with a 20-thread CPU.

References

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[12] Persistence research repository, `https://github.com/CorvusCorpus/persistence-research` (`docs/CHAIN_SIEVE.md`, `docs/NOTEBOOK.md`).